

Career Progression and Promotion Gap Analysis for Retention Optimization at Palo Alto Networks

1. Abstract

Employee retention can be influenced by several career-related factors, including opportunities for promotion, time spent in the same role, training exposure, and managerial continuity. Traditional employee attrition analysis generally focuses on predicting whether an employee may leave, but such analysis may not provide sufficient insight into the underlying career progression patterns that can contribute to employee disengagement.

This project, “**Career Progression and Promotion Gap Analysis for Retention Optimization at Palo Alto Networks,**” focuses on analyzing employee career trajectories and identifying potential career progression gaps. An HR employee dataset containing demographic, job, compensation, career, satisfaction, and attrition-related information was analyzed.

Several career progression features were engineered, including **Promotion Gap Ratio, Role Stagnation Index, Training Intensity Score, Manager Stability Indicator, Career Velocity Score, Salary Growth Proxy, and Career Stage**. Exploratory data analysis was performed to understand promotion gaps, role stagnation, attrition patterns, and relationships between career-related variables.

K-Means clustering was then applied to standardized career progression features to identify groups of employees with similar career characteristics. Four career clusters were generated, while hierarchical clustering was used as an additional validation approach.

A project-defined **Promotion Gap Risk Score, Retention Opportunity Index, and Training Need Indicator** were subsequently developed to support structured analysis of career development opportunities.

The resulting analysis was integrated into an interactive **Streamlit dashboard** containing filters, key performance indicators, employee-level career metrics, cluster analysis, promotion gap analysis, training needs, manager stability analysis, and retention opportunity analysis.

The project provides an analytical framework for examining career progression patterns and identifying areas where employee development and retention initiatives may be considered.

2. Introduction

Employee retention is an important organizational challenge because employee turnover can result in recruitment costs, knowledge loss, productivity disruption, and additional workload for existing employees.

While attrition analysis can identify historical patterns associated with employees leaving an organization, understanding **why career progression may differ across employees** can provide additional context.

Employees may experience different career trajectories depending on:

- time spent within an organization,
- time spent in their current role,
- time since their last promotion,
- training exposure,
- continuity with their current manager,
- job level, and
- overall career progression.

The objective of this project is therefore not simply to identify employees who historically left the organization. Instead, it examines employee career progression patterns and develops analytical indicators that can help identify potential areas for career development and retention intervention.

3. Problem Statement

Traditional HR attrition analysis often concentrates on identifying employees who are likely to leave. However, such analysis may not sufficiently explain **career progression gaps** that exist within an employee population.

Employees with long periods in the same role, limited promotion progression, relatively low training exposure, or changing managerial relationships may have different career experiences.

Therefore, the problem addressed in this project is:

How can employee career progression data be analyzed to identify career clusters, promotion gaps, role stagnation, training needs, managerial stability patterns, and potential retention opportunities?

The project uses exploratory analysis, feature engineering, clustering, and rule-based analytical indicators to address this problem.

4. Project Objectives

The main objectives of the project are:

- Analyze employee career progression patterns.
- Identify potential promotion gaps.
- Measure role stagnation using employee career data.
- Analyze training intensity in relation to employee tenure.
- Examine managerial continuity.
- Group employees into career progression clusters.
- Develop a Promotion Gap Risk Score.
- Develop a Retention Opportunity Index.
- Identify potential employee development needs.
- Analyze career progression patterns across departments, job roles, and career stages.
- Develop an interactive Streamlit dashboard for HR analysis.
- Provide structured management insights based on the analytical results.

5. Dataset Description

The project uses an employee HR dataset associated with Palo Alto Networks.

The dataset contains **1,470 employee records**.

Important variables include:

Category	Variables
Employee information	Age, Gender, MaritalStatus
Organization	Department, JobRole
Career	JobLevel, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion
Management	YearsWithCurrManager
Training	TrainingTimesLastYear
Compensation	MonthlyIncome, PercentSalaryHike
Work experience	TotalWorkingYears, NumCompaniesWorked
Satisfaction	JobSatisfaction, EnvironmentSatisfaction, RelationshipSatisfaction
Work conditions	OverTime, BusinessTravel, WorkLifeBalance
Outcome	Attrition

The target variable **Attrition** contains historical employee attrition information.

6. Data Preparation

The dataset was initially inspected for:

- dataset dimensions,
- missing values,
- duplicate records,
- data types, and
- basic statistical characteristics.

Duplicate records and missing-value checks were performed before beginning the analytical process.

The dataset contained **1,470 employee records**, which were used as the basis for the career progression analysis.

7. Feature Engineering

Feature engineering was one of the main parts of the project.

Several new variables were created to represent different aspects of employee career progression.

7.1 Promotion Gap Ratio

The Promotion Gap Ratio measures the proportion of an employee's time at the company represented by the period since their last promotion.

$$PromotionGapRatio = \frac{YearsSinceLastPromotion}{YearsAtCompany}$$

To avoid division by zero, a safe version of YearsAtCompany was used when the value was zero.

A higher ratio indicates that a larger proportion of the employees' organizational tenure has occurred since their most recent promotion.

7.2 Role Stagnation Index

The Role Stagnation Index measures how much of an employee's company tenure has been spent in the current role.

$$RoleStagnationIndex = \frac{YearsInCurrentRole}{YearsAtCompany}$$

A higher value indicates that the employee has spent a larger proportion of their company tenure in their current role.

7.3 Training Intensity Score

Training intensity was normalized relative to company tenure.

$$TrainingIntensityScore = \frac{TrainingTimesLastYear}{YearsAtCompany}$$

This provides a relative measure of training exposure rather than using the raw number of training sessions alone.

7.4 Manager Stability Indicator

Managerial continuity was represented using:

$$ManagerStabilityIndicator = \frac{YearsWithCurrManager}{YearsAtCompany}$$

A higher value indicates that a larger proportion of the employee's company tenure has been spent with the current manager.

7.5 Career Velocity Score

A Career Velocity Score was created using job level relative to total working experience:

$$CareerVelocityScore = \frac{JobLevel}{TotalWorkingYears}$$

Safe versions of the denominator were used for zero values.

This variable provides a simplified indicator of career progression relative to total experience.

7.6 Salary Growth Proxy

A simple salary-related proxy was also created:

$$SalaryGrowthProxy = PercentSalaryHike \times JobLevel$$

This variable was retained for employee-level analysis and profiling rather than being used as a primary clustering feature.

7.7 Career Stage

Employees were grouped into custom career stages based on years at the company:

Years at Company	Career Stage
Less than 3	Early Career
3–6	Growth Stage
7–14	Mid Career
15 or more	Senior Career

These thresholds were defined specifically for this project to support career-stage analysis.

8. Exploratory Data Analysis

Exploratory data analysis was performed before clustering to understand the underlying employee career patterns.

8.1 Attrition Distribution

Historical attrition was examined to understand the distribution of employees who remained versus employees who left.

The dataset contains approximately:

- **83.9% historical non-attrition**
- **16.1% historical attrition**

This provides a baseline for subsequent career progression analysis.

8.2 Years Since Last Promotion

The distribution of YearsSinceLastPromotion was analyzed.

The analysis showed that many employees had relatively recent promotions, while a smaller group had substantially longer periods since their last promotion.

This makes the variable useful for investigating potential promotion gaps.

8.3 Years in Current Role

YearsInCurrentRole was analyzed to understand how long employees typically remained in their current positions.

The distribution provided insight into role tenure and helped motivate the creation of the **Role Stagnation Index**.

8.4 Promotion Gap Ratio

The Promotion Gap Ratio was examined through distribution analysis and department-level comparisons.

The analysis showed a concentration of employees with relatively low ratios, along with employees exhibiting substantially higher ratios.

Department-level analysis was also performed to compare career progression patterns.

8.5 Promotion Gap and Historical Attrition

Promotion Gap Ratio was compared against historical attrition.

This analysis was used to examine whether employees with different promotion-gap characteristics also displayed differences in historical attrition patterns.

Importantly, this analysis describes historical associations in the dataset and does not establish that promotion gaps cause attrition.

8.6 Role Stagnation by Job Role

The Role Stagnation Index was analyzed across job roles.

This allowed comparison of role progression patterns between different types of positions.

8.7 Correlation Analysis

A correlation heatmap was created using the career progression features.

The analysis included:

- Promotion Gap Ratio
- Role Stagnation Index
- Training Intensity Score
- Manager Stability Indicator
- Career Velocity Score
- Years at Company
- Years in Current Role
- Years Since Last Promotion
- Years with Current Manager
- Attrition

Attrition was included for exploratory analysis but **was not used as a feature for K-Means clustering.**

9. Clustering Methodology

The next stage of the project was to identify employee career progression groups.

9.1 Clustering Features

The following nine features were used for clustering:

1. PromotionGapRatio
2. RoleStagnationIndex
3. TrainingIntensityScore
4. ManagerStabilityIndicator
5. CareerVelocityScore
6. YearsAtCompany
7. YearsInCurrentRole
8. YearsSinceLastPromotion
9. YearsWithCurrManager

Historical Attrition was deliberately excluded from the clustering features.

This prevents the clustering process from directly using the outcome variable to define employee groups.

10. Data Scaling

Because the clustering variables have different numerical ranges, **StandardScaler** was used.

Standardization transforms the variables so that they have comparable scales.

This is important for K-Means because distance calculations can otherwise be dominated by variables with larger numerical ranges.

11. Selecting the Number of Clusters

The **Elbow Method** was used to examine different values of K.

Values between 2 and 10 clusters were evaluated.

A **working value of K = 4** was selected based on the elbow analysis and interpretability of the resulting employee groups.

Silhouette scores were also calculated as an additional clustering diagnostic.

The project does not treat K=4 as a universally optimal solution; rather, it was selected as an interpretable clustering configuration for this analysis.

12. K-Means Clustering

K-Means clustering was implemented using:

- `n_clusters = 4`
- `random_state = 42`
- `n_init = 10`

The resulting dataset contained four career clusters. The distribution was:

Career Cluster	Employees
Cluster 0	728
Cluster 1	209
Cluster 2	294
Cluster 3	239
Total	1,470

The clusters were then profiled using average career progression characteristics.

13. Career Cluster Interpretation

The clusters were interpreted using differences in:

- promotion gap,
- role stagnation,
- training intensity,
- manager stability,
- career velocity,
- company tenure,
- current-role tenure,
- time since promotion, and
- manager tenure.

The purpose of cluster interpretation was to identify different **career progression patterns**, rather than assigning permanent labels to individual employees.

Possible interpretations include patterns such as:

- employees with faster career progression,
- stable long-term contributors,
- employees with relatively shorter organizational tenure,
- employees with stronger indicators of career stagnation.

The exact characteristics of each cluster should be interpreted from the generated cluster profile rather than assuming that a cluster number itself has an inherent meaning.

14. Hierarchical Clustering Validation

Hierarchical clustering using **Agglomerative Clustering** with Ward linkage was also performed using four clusters.

The resulting hierarchical cluster assignments were compared with the K-Means assignments using:

- cross-tabulation, and
- Adjusted Rand Index (ARI).

This provided an additional method for assessing whether similar grouping structures appeared under a different clustering algorithm.

The hierarchical clustering was used as a validation/comparison exercise rather than replacing K-Means as the primary clustering method.

15. Promotion Gap Risk Score

A project-defined Promotion Gap Risk Score was developed to combine two career progression indicators:

$$\begin{aligned} PromotionGapRiskScore \\ = 0.6(PromotionGapRatio) + 0.4(RoleStagnationIndex) \end{aligned}$$

The score gives greater weight to the Promotion Gap Ratio while also considering role stagnation.

Employees were divided into three categories using dataset quantiles:

- **Low**
- **Medium**
- **High**

The 33rd and 67th percentiles were used as the boundaries.

Important interpretation

This is a **project-defined analytical indicator**, not a clinically, statistically, or organizationally validated risk model.

A high score indicates stronger career progression-gap characteristics within this project's framework.

16. Retention Opportunity Index

A Retention Opportunity Index was developed to identify employees exhibiting multiple career-development characteristics.

The following normalized variables were used:

- Role Stagnation
- Training Intensity
- Manager Stability

The project-defined formula was:

$$\begin{aligned} \textit{RetentionOpportunityIndex} \\ &= 0.50(\textit{PromotionGapRiskScore}) + 0.30(\textit{RoleStagnationNorm}) \\ &+ 0.20(1 - \textit{TrainingIntensityNorm}) \end{aligned}$$

The resulting score was divided into:

- Low
- Medium
- High

using the 33rd and 67th percentiles.

The purpose is to identify employees whose career characteristics may warrant closer examination for development or retention initiatives.

It should not be interpreted as a prediction that an employee will leave.

17. Training Need Indicator

Training exposure was analyzed using the Training Intensity Score.

The median Training Intensity Score was used as a project-defined threshold.

Employees below the median were categorized as:

Needs Development

Employees at or above the median were categorized as:

Adequate

This indicator is intended to support analysis of relative training exposure.

18. Manager Stability Analysis

Managerial continuity was examined using the Manager Stability Indicator.

The median indicator value was used to categorize employees into:

- **Lower Stability**
- **Higher Stability**

Manager stability was subsequently compared with promotion-gap risk.

This provides a way to examine whether career progression characteristics differ across levels of managerial continuity.

However, manager stability should not automatically be interpreted as positive or negative without organizational context.

19. Retention Opportunity Analysis

A subset of employees was identified using the following conditions:

- historical Attrition = No
- PromotionGapRisk = High
- RetentionOpportunity = High

This produced a group representing employees who historically remained with the organization but also exhibit high career-development indicators under the project's framework.

This group was analyzed by:

- department,
- job role, and
- career stage.

Important limitation

Because the dataset records historical attrition, Attrition = No means the employee did not leave during the period represented by the dataset. It does **not** mean the employee is currently employed or will remain employed in the future.

20. Department-Level Analysis

Career progression and retention indicators were analyzed across departments.

The department-level analysis included:

- employee count,
- average Promotion Gap Risk Score,
- average Role Stagnation Index,
- average Training Intensity,
- average Manager Stability,
- number of employees with high promotion-gap risk,
- number of high retention-opportunity employees,
- number of training needs, and
- historical attrition rate.

This provides a department-level view of career progression characteristics.

21. Career Stage Analysis

Career stages were used to compare career progression patterns across:

- Early Career
- Growth Stage
- Mid-Career
- Senior Career

High retention opportunities were also analyzed across these career stages.

This helps identify whether development opportunities are concentrated in particular stages of an employee's organizational journey.

22. Key Performance Indicators

The final employee-level KPI dataset contains several analytical indicators.

Important KPIs include:

KPI	Purpose
Career Cluster	Identifies employee career progression group
Promotion Gap Risk Score	Measures project-defined promotion-gap characteristics
Promotion Gap Risk	Low/Medium/High classification
Retention Opportunity Index	Identifies potential development/retention opportunities
Retention Opportunity	Low/Medium/High classification
Training Need Indicator	Identifies relative training-development need
Manager Stability	Relative manager continuity category
Promotion Gap Ratio	Measures promotion gap relative to company tenure
Role Stagnation Index	Measures current-role tenure relative to company tenure
Training Intensity Score	Measures training exposure relative to company tenure
Manager Stability Indicator	Measures manager continuity relative to company tenure
Career Velocity Score	Career progression proxy
Career Stage	Employee career-stage classification

23. Streamlit Dashboard

An interactive Streamlit dashboard was developed to make the analysis easier to explore.

The dashboard uses the final employee KPI dataset exported from the analysis.

Dashboard Filters

Users can filter employees by:

- Department
- Job Role
- Career Stage
- Promotion Gap Risk

All major dashboard visualizations update according to the selected filters.

24. Dashboard KPIs

The dashboard displays:

Total Employees

Shows the number of employees remaining after applying the selected filters.

Historical Attrition

Displays the historical attrition percentage for the filtered population.

High Promotion Gap Risk

Shows the number of employees categorized as having high Promotion Gap Risk.

High Retention Opportunities

Shows the number of employees categorized as having high Retention Opportunity.

Training Needs

Shows the number of employees classified as needing development.

For the complete unfiltered dataset, the dashboard showed:

- **1,470 employees**
 - **16.1% historical attrition**
 - **462 high promotion-gap-risk employees**
 - **485 high retention-opportunity employees**
 - **713 employees needing development**
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25. Dashboard Visualizations

The dashboard contains the following visualizations:

25.1 Career Cluster Distribution

Displays the number of employees in each career cluster.

25.2 Promotion Gap Risk Distribution

Shows the distribution of employees across Low, Medium, and High promotion-gap-risk categories.

25.3 Historical Attrition Rate by Career Cluster

Compares historical attrition rates across the four career clusters.

25.4 High Retention Opportunities by Department

Shows the number of high retention-opportunity employees in each department.

25.5 Training Needs by Department

Shows the distribution of training-development needs across departments.

25.6 Manager Stability vs Promotion Gap Risk

Examine the relationship between manager stability categories and promotion-gap risk.

25.7 Career Cluster Profile

A heatmap compares average career progression characteristics across clusters.

25.8 High Retention Opportunities by Career Stage

Shows high retention-opportunity employees across the defined career stages.

26. Management Insights

The analysis can support several areas of HR and management review.

Career Progression

Employees exhibiting relatively high promotion-gap or role-stagnation indicators can be reviewed for career-development opportunities.

Training

Employees classified as needing development can be examined to determine whether additional training or skill-development opportunities are appropriate.

Managerial Continuity

Manager stability indicators can provide additional context when examining employee career experiences.

Career Stage

Different career stages may require different development approaches.

For example, early-career employees may require skill development and role exposure, while more experienced employees may require broader career progression opportunities.

These are potential management considerations derived from the analytical framework rather than automatic recommendations for individual employees.

27. Project Findings

The completed analysis provides several broad findings.

Finding 1 — Career progression varies across employees

Employees display substantially different combinations of promotion gaps, role tenure, training exposure, and managerial continuity.

This supports the use of employee segmentation rather than treating the workforce as a single homogeneous population.

Finding 2 — Promotion gap and role stagnation provide complementary information

Years since promotion alone does not fully represent career stagnation.

Combining Promotion Gap Ratio with Role Stagnation Index provides a broader view of career progression.

Finding 3 — Career clusters provide a structured segmentation

The K-Means analysis produced four employee groups with different career progression characteristics.

These groups allow HR analysts to examine career patterns at a segment level.

Finding 4 — Training exposure can be examined alongside career progression

Training Intensity Score provides additional context when analyzing employees with higher career stagnation indicators.

Finding 5 — Managerial continuity provides another dimension of career analysis

The Manager Stability Indicator can be used to examine whether employees experience different career progression patterns based on managerial continuity.

Finding 6 — Retention opportunity can be separated from historical attrition

The project does not rely only on historical attrition.

It also identifies employees who remained in the historical dataset but exhibit career-development characteristics that may warrant attention.

28. Business Value

The project can support HR teams by providing a structured way to:

- monitor career progression,
- identify potential promotion gaps,
- segment employees based on career characteristics,
- examine training-development needs,
- analyze managerial continuity,
- explore department-level patterns, and
- identify potential retention-development opportunities.

Instead of viewing employee retention only as an attrition prediction problem, the project introduces a **career progression perspective**.

29. Limitations

Several limitations should be considered.

29.1 Historical Dataset

The dataset represents historical employee information. Therefore, the results should not automatically be interpreted as current employee conditions.

29.2 Project-Defined Scores

Promotion Gap Risk Score and Retention Opportunity Index were designed specifically for this project.

They have not been independently validated against future employee outcomes.

29.3 No Causal Inference

The analysis identifies associations and patterns.

It does not establish that promotion gaps, manager stability, training, or role stagnation directly cause attrition.

29.4 Outliers

Potential outliers were identified using the IQR method during preprocessing.

However, they were **not removed or capped** in the final analysis.

29.5 Clustering Limitations

K-Means clustering depends on feature selection, scaling, and the selected number of clusters.

The four-cluster solution should therefore be interpreted as an analytical segmentation rather than an objectively permanent classification of employees.

29.6 Categorical Variables

Department and JobRole were primarily used for profiling and analysis.

The K-Means model itself was based on numeric career progression features rather than directly encoding categorical variables.

29.7 Career Stage Thresholds

Career stages were defined using project-specific thresholds and may not correspond to official organizational career frameworks.

30. Future Scope

The project can be extended in several ways.

30.1 Predictive Attrition Modeling

A future version could develop a separate supervised machine-learning model to predict attrition.

Possible models include:

- Logistic Regression
- Random Forest
- XGBoost
- Gradient Boosting

This should be kept separate from the career clustering model.

30.2 Time-Series Career Analysis

If multiple years of employee records were available, employee career trajectories could be studied over time rather than using a single historical snapshot.

30.3 Promotion Prediction

A future model could estimate the probability of an employee receiving a promotion based on historical career progression patterns.

30.4 Personalized Development Recommendations

The dashboard could eventually provide structured development suggestions based on:

- career cluster,
- role,
- training exposure,
- promotion gap,
- career stage, and
- skill requirements.

30.5 Automated HR Alerts

The dashboard could be extended with alerts for predefined career progression thresholds.

30.6 Model Validation

The project-defined risk and retention indicators could be validated against future employee outcomes when suitable longitudinal data becomes available.

30.7 Explainable AI

Explainability techniques could be introduced if predictive models are added in the future, allowing HR users to understand which variables contribute to predictions.

31. Technology Stack

The project was implemented using:

Programming Language

- Python

Data Analysis

- Pandas
- NumPy

Visualization

- Matplotlib
- Seaborn
- Plotly

Machine Learning

- Scikit-learn
- K-Means Clustering
- Agglomerative Clustering
- StandardScaler
- MinMaxScaler
- Silhouette Score
- Adjusted Rand Index

Dashboard

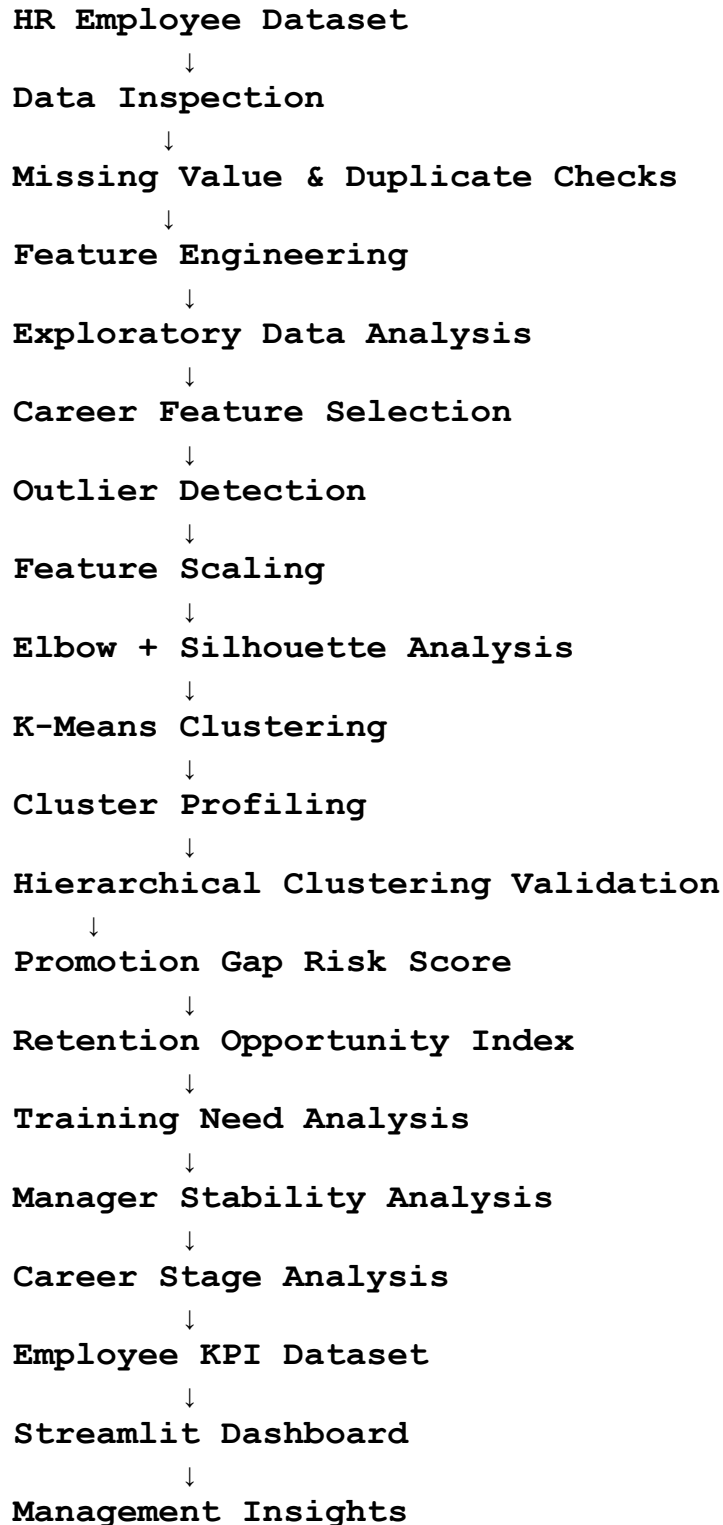
- Streamlit

Development Environment

- Google Colab
- Visual Studio Code

32. Project Workflow

The overall project workflow can be summarized as:



33. Conclusion

This project demonstrates how HR employee data can be used to analyze career progression patterns beyond traditional attrition analysis.

By engineering career-focused indicators such as **Promotion Gap Ratio, Role Stagnation Index, Training Intensity Score, Manager Stability Indicator, and Career Velocity Score**, the project provides multiple perspectives on employee career experiences.

K-Means clustering was used to segment employees into four career progression groups, while hierarchical clustering provided an additional comparison of the employee grouping structure. Additional project-defined indicators were then developed to identify promotion-gap characteristics, potential retention opportunities, training-development needs, and managerial stability patterns.

The Streamlit dashboard converts the analytical results into an interactive interface where users can filter employees and explore career clusters, promotion risks, training needs, manager stability, career stages, and retention opportunities.

Overall, the project provides a **career progression intelligence framework** that can complement traditional HR attrition analysis. Its primary value is in helping analysts and management explore where career-development attention may be warranted, while recognizing that the project's risk and opportunity indicators require further validation before being used as formal HR decision-making models.

34. Final Project Deliverables

Analysis

- EDA
- Feature engineering
- Career progression analysis
- K-Means clustering
- Hierarchical clustering comparison
- Promotion Gap Risk analysis
- Retention Opportunity analysis
- Training Need analysis
- Manager Stability analysis
- Career Stage analysis

Exported datasets

- Palo_Alto_Career_Progression_Analysis.csv
- Career_Cluster_Profile.csv
- Department_Management_Insights.csv
- Final_Employee_Career_KPI_Dataset.csv

Dashboard

- app.py
 - Final_Employee_Career_KPI_Dataset.csv
 - Interactive Streamlit dashboard
-